

Towards a Decentralized Sphinx Header Construction

Aurélien Chassagne

August 12, 2025

① Introduction

- Motivation - Speak about importance of Privacy on the network
- Some Technologies: TOR vs Mixnet (lot of animations)

② The problem

- Mixnet's problem (discard illegal packet at exit gateway)
- Our idea - Decentralization

③ Sphinx

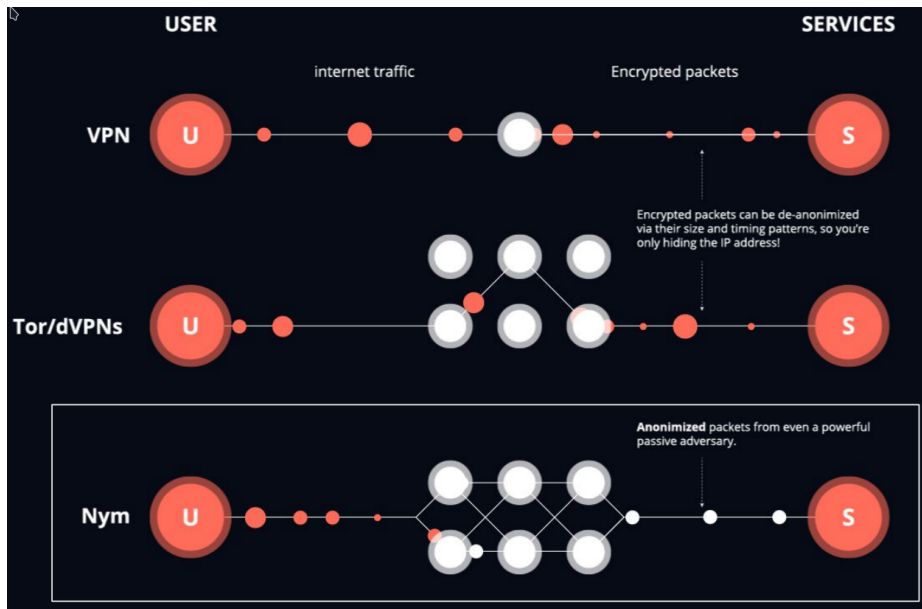
- Original Sphinx (lot of animations)
- Our Sphinx implementation (lot of animations)

④ Tests and Results

⑤ Conclusion and future work

– Why privacy matters –

Network Privacy Technologies



The Problem - Policy enforcement at exit gateway

Why *policy enforcement*:

- ① **Economic:** Company willing to provide different subscription plans
- ② **Political:** Enforce law regulation (i.e. reaching illegal website)

Why at *exit gateway*:

- ① Privacy-friendly legitimacy verification of layered encrypted packet is tricky
- ② At exit gateway, the destination is revealed allowing easier policy enforcement

Why is it a *problem*:

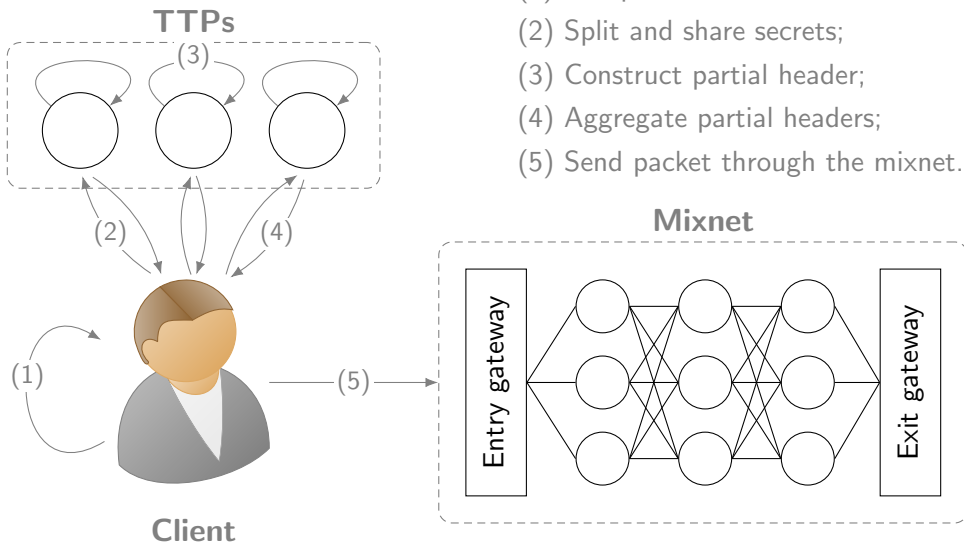
- ① Computation waste of mixnodes

What we would like:

- Policy enforcement applicable at any nodes

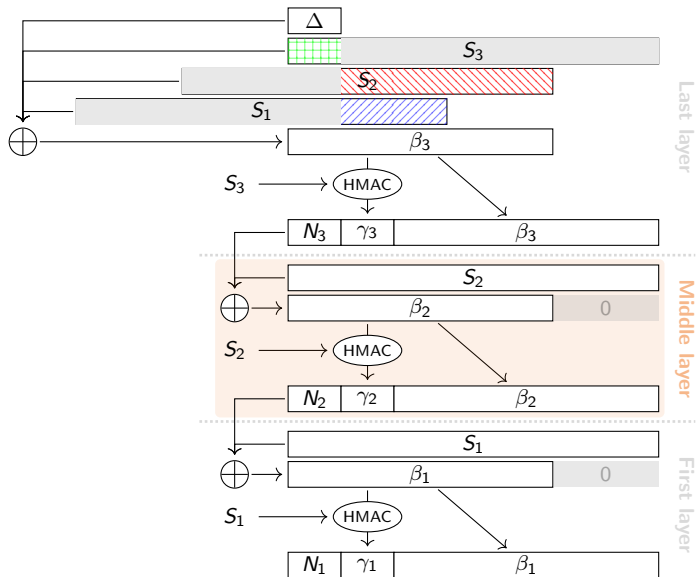
Our Idea - Decentralized packet construction

- (1) Compute route & shared secrets;
- (2) Split and share secrets;
- (3) Construct partial header;
- (4) Aggregate partial headers;
- (5) Send packet through the mixnet.



- Upper part: Mixnet path
- Middle part: Packet representation
- Lower part: Packet processing (Fig 4)

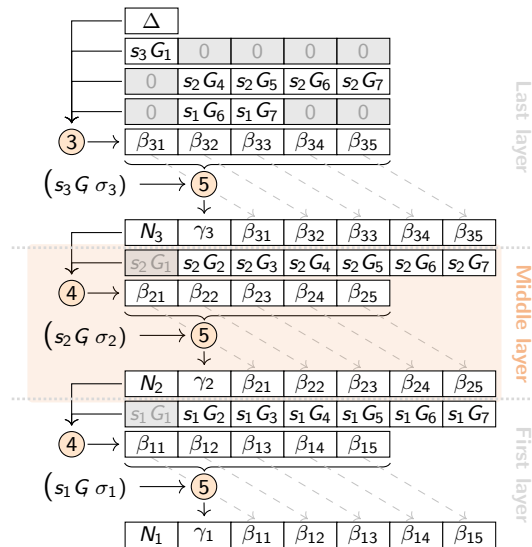
Original Sphinx - Encryption



ECC slide

Is it needed ? Generator ?

Our decentralized Sphinx construction



- Complexity - Big O notation
- Unlinkability
 - NIST - P-value distribution (Fig 6 → Fig 7)
 - NIST - Proportion of success (Fig 5)
- Recap table (Tab. 2)

Conclusion and Future Work